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DATA 3421

Final Project Report: Titanic Dataset

Introduction

This final project report was conducted by Tej Govindu and Mateen Jan for DATA 3421: DATA MINING MGT CURATION, instructed by Dr. Qiang Zheng.

1. Statement of the Problem

The project uses Titanic dataset analysis to research which factors most affected passenger survival rates during the RMS Titanic disaster. The dataset provides complete passenger information which includes their demographic data and socioeconomic details and their travel-related information during their trip. The project will use various different data mining techniques learned throughout the semester to find complex patterns in the dataset which will create predictive models for passenger survival prediction and outcomes.

The problem provides historical significance and analytical value by enabling practical dataset classification through various classification methods. We determined which factors, including gender and age and passenger class and fare, had the strongest effect on survival outcomes during disembarkation. The project evaluates machine learning models ability to forecast survival based on their specific features.

To guide our analysis, we will focus on the following key questions:

- Which features (gender, age, passenger class, fare) had the strongest impact on survival?
- Were certain groups (women, children, or first-class passengers) more likely to survive?
- Can we accurately predict whether a passenger survived using the available features?

The research questions we address will help us identify which variables have the greatest impact on survival rates. The analysis creates a dual value system because it uses historical data and statistical methods to produce insights from unprocessed data. The project objectives establish two primary targets which involve dataset comprehension and demonstration of mining methods that solve classification challenges through data-driven decision support.

2. Data

The dataset used in this project was obtained from Kaggle and contains information about passengers from the Titanic. It includes approximately 891 observations with passenger details.

Variables Used:

- Survived (target variable): Indicates whether a passenger survived (1) or not (0)
- Pclass: Passenger class (1st, 2nd, 3rd)
- Sex: Gender of the passenger
- Age: Age in years
- SibSp: Number of siblings/spouses aboard
- Parch: Number of parents/children aboard
- Fare: Ticket fare
- Embarked: Port of embarkation

3. Analyses and Modeling

Data Preprocessing:

Before conducting any analysis, the dataset was cleaned and transformed. Missing values in the Age column were replaced with the median value, while missing values in Embarked were filled using the mode. The Cabin column was removed due to excessive missing data. Categorical variables were encoded, and irrelevant columns were removed.

Exploratory Data Analysis:

Exploratory data analysis was conducted to identify patterns and relationships within the dataset. The overall survival distribution is shown in Figure 1, which highlights the imbalance between survivors and non-survivors.

Analysis by gender revealed a strong relationship between gender and survival, as illustrated in Figure 2.

The passenger class also played a significant role in survival outcomes. As shown in Figure 3, first-class passengers had higher survival rates compared to those in lower classes.

The age distribution of passengers is presented in Figure 4, which provides insight into the demographic spread of individuals aboard the Titanic.

To further understand relationships between variables, a correlation heatmap was generated, shown in Figure 5.

Model Building:

Multiple classification models were used to develop survival prediction. We used three models which were Logistic Regression and Decision Tree and Random Forest to conduct research. Two sets of data were built which included training data and testing data to assess model performance.

Model Evaluation:

We used accuracy and confusion matrices and classification reports to assess model performance.

We assessed each model performance through evaluation while conducting a comparison between models. Figure 6 displays the complete comparison which includes all three models.

The visual representation shows how different models perform by displaying their accuracies.

We created a combined feature importance comparison to study how each model interprets feature significance. The process enables the establishment of valid basis for comparison since all models use different methods to calculate importance values. The analysis results are displayed in Figure 7. The visual representation enables viewers to understand which variables usually predict outcomes throughout various modeling methods.

4. Results

The results provide clear insights into both model performance and the factors influencing passenger survival. The Random Forest model achieved the highest accuracy between the three models while Logistic Regression established a strong baseline and Decision Tree performed slightly lower.

The feature importance comparison in Figure 7 shows that gender and passenger class were the most influential variables across all models. Other features contributed but had less impact. The models show different importance levels because of their different methods which enable Logistic Regression to discover linear patterns while tree models detect relationships.

5. Conclusions

This project successfully applied mining techniques to analyze the Titanic dataset and answer questions. The findings show that gender and passenger class were the most influential survival factors. The predictive models developed were effective in classifying survival outcomes.

6. Discussion

While the project produced critical results, several limitations should be noted. Missing data required handling, which may introduce some bias when drawing conclusions.

Future improvements could include:

- Advanced models such as gradient boosting
- Feature engineering (e.g., family size)
- Cross-validation for more robust evaluation

7. References

- Kaggle Titanic Dataset: <https://www.kaggle.com/datasets/yasserh/titanic-dataset>
- Python libraries: pandas, sklearn, matplotlib, seaborn

8. Individual Contribution Reports

Mateen:

My partner and I used the Titanic dataset for our project. My partner and I worked together to clean the data and handle missing values. He and I created graphs while testing different ML models. I assisted in the analysis of survival patterns which showed results based on gender, class, age, and fare. This project enabled me to develop my skills in programming, visualization techniques, and machine learning applications.

Tej:

I worked with my partner to analyze the Titanic dataset. I helped clean the data, handle missing values, and prepare it for analysis. My work included creating visualizations and developing ML models such as the feature importances and confusion matrices. I helped interpret results while comparing model performance through various testing methods.

Appendix

** This section does not exceed the maximum 5-page limit for the Report. It is meant to showcase the Figure images listed in the Report itself, making it easier to understand. **

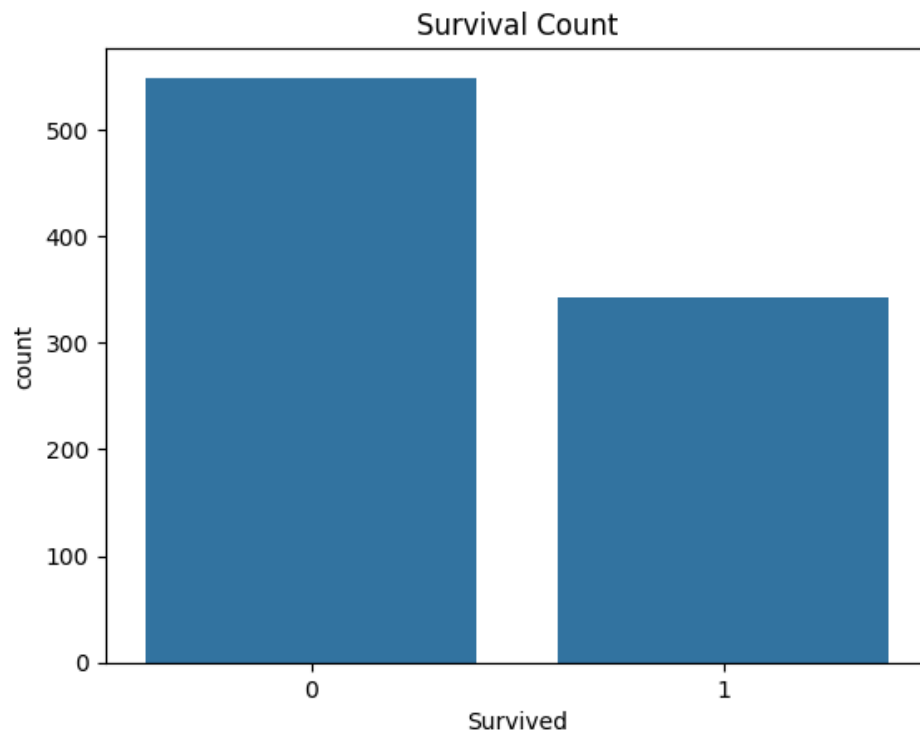


Figure 1:

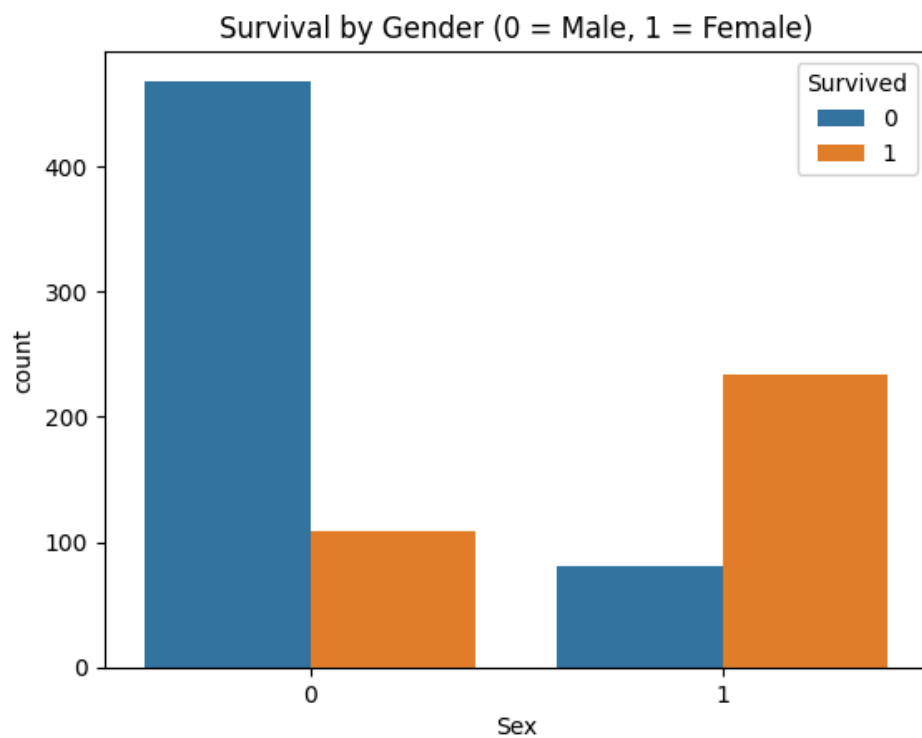


Figure 2:

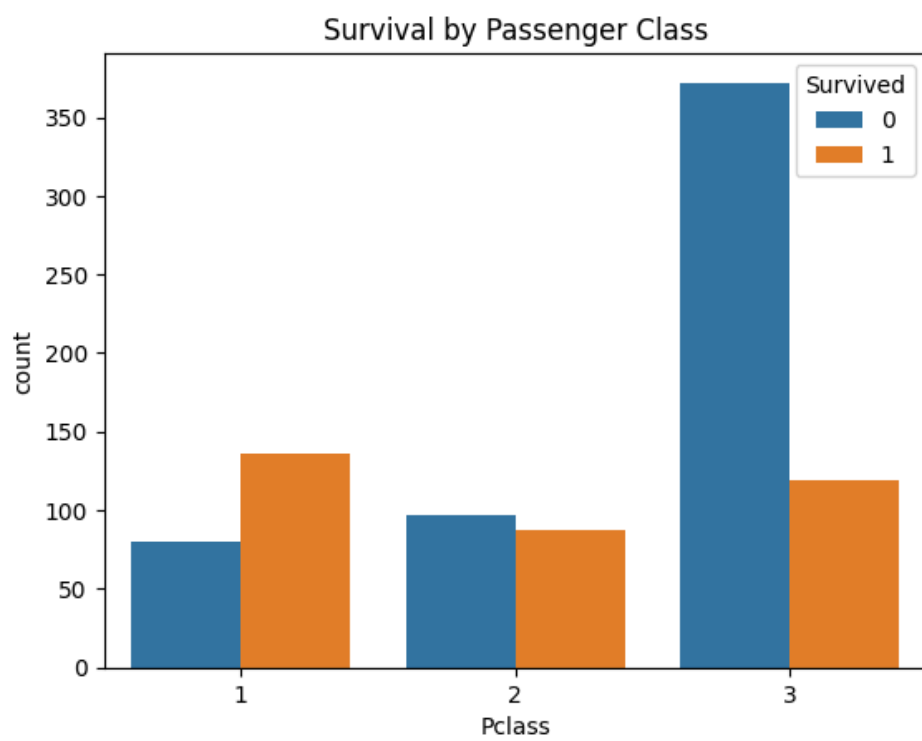


Figure 3:

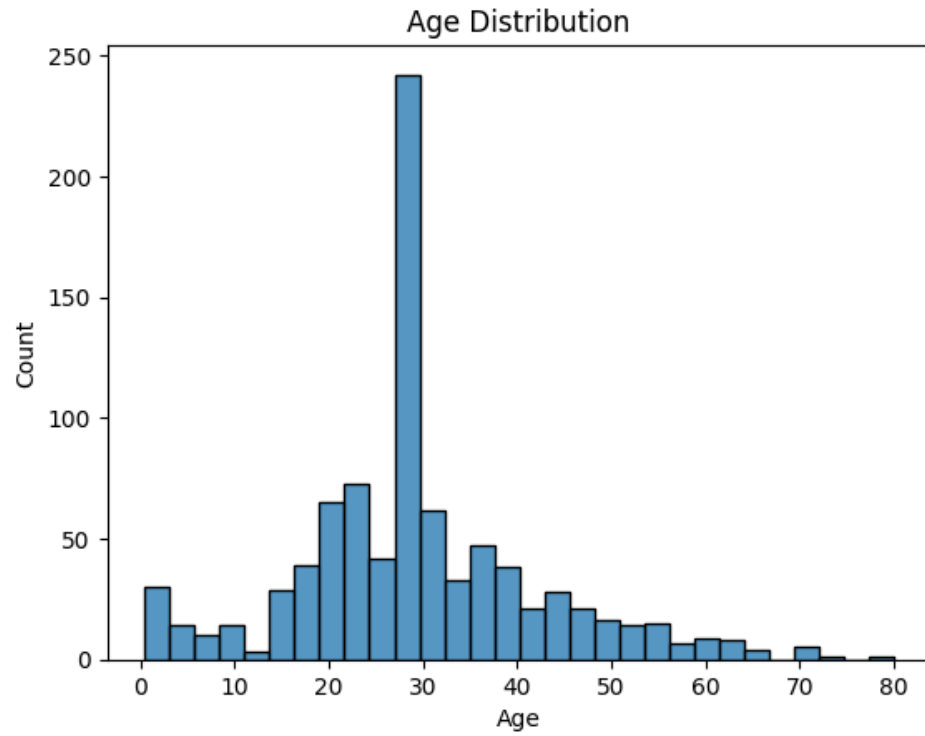


Figure 4:

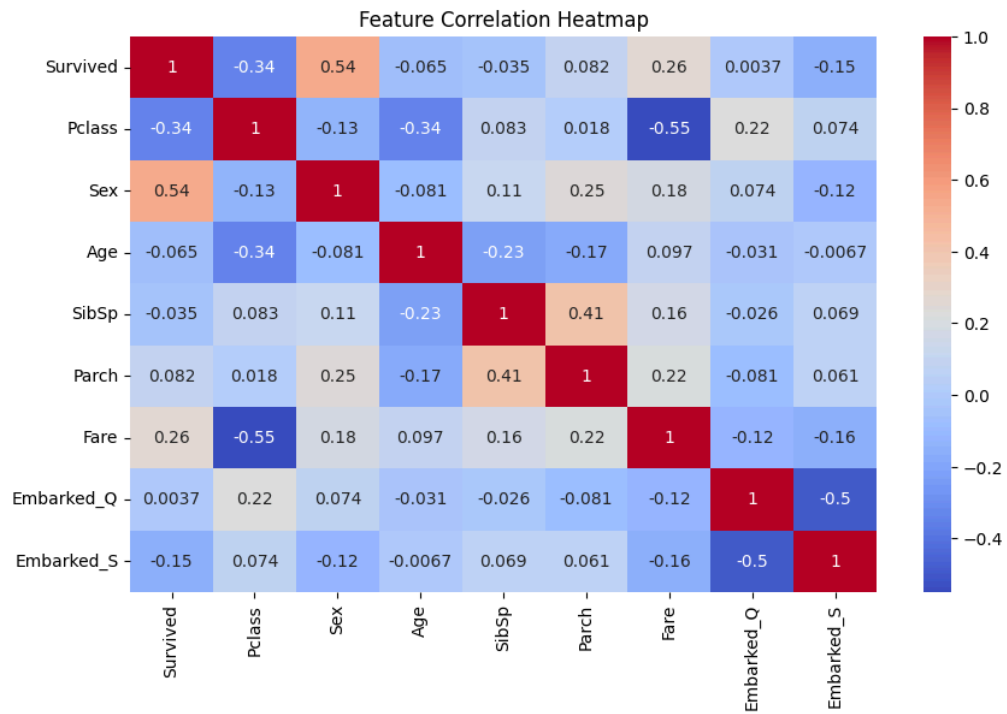


Figure 5:

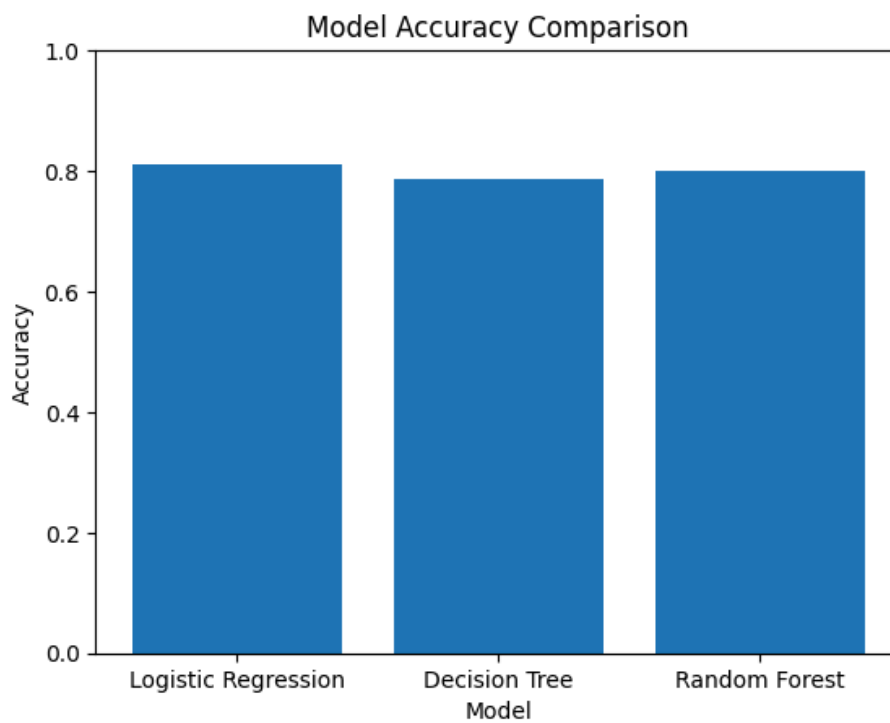


Figure 6:

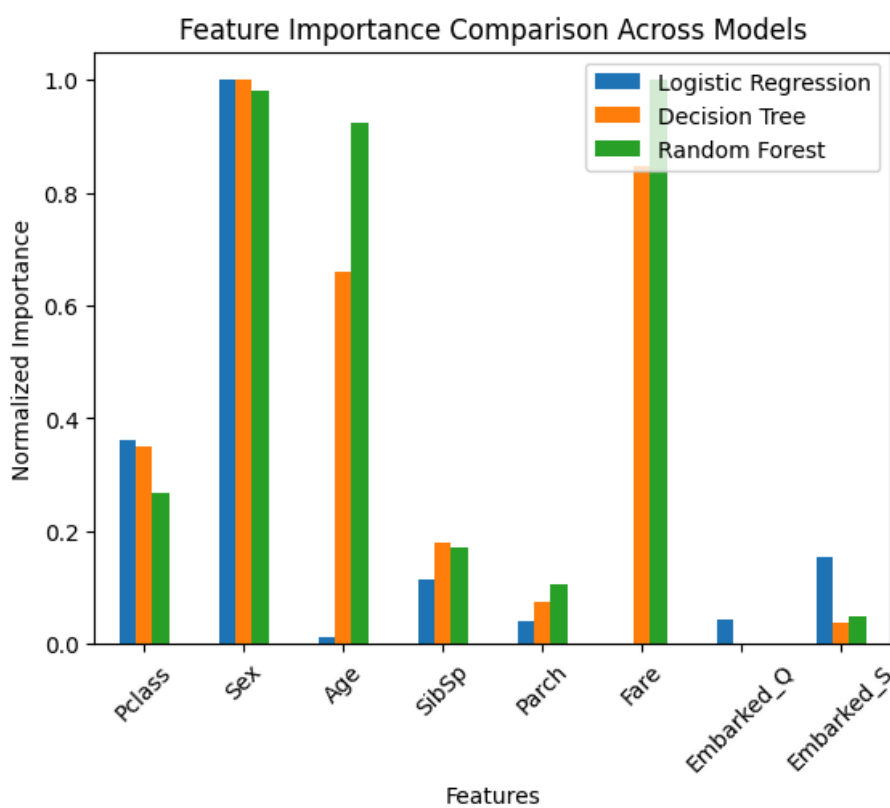


Figure 7:

Additional Figures (Not mentioned) -

